JAIRAJ BANDA

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OBJECTIVE

Final-year B.Tech ECE student with hands-on experience building embedded and robotics prototypes from scratch — sensors, motor control, and microcontroller-driven systems. Comfortable researching and selecting components with limited direction, iterating through repeated testing, and documenting results. Seeking a hands-on hardware/robotics internship to help build 0-to-1 experimental prototypes.

TECHNICAL SKILLS

Languages: C, C++, Embedded C, Python (basic–intermediate), Verilog HDL
Microcontrollers: ESP32, STM32 (BlackPill F411CEU6), Arduino Uno
Sensors & Actuators: Ultrasonic sensors, servo motors, relay modules, DC/BLDC motor drivers (TB6612FNG, DRV8833), ESCs
Protocols & Peripherals: UART, I2C, SPI, CAN, PWM/timer configuration, GPIO/memory-mapped I/O
Tools & Platforms: Arduino IDE, STM32CubeIDE/CubeMX, Xilinx Vivado, Blynk IoT, VS Code, EDA Playground

HARDWARE & ROBOTICS PROJECTS

ESP32-Based Radar System — *Embedded Systems / Sensors & Actuators* | ESP32, Ultrasonic sensor, Servo motor, Arduino IDE

- Designed and assembled a 180° sweep radar prototype combining a servo motor and ultrasonic sensor to detect nearby objects.
- Wrote embedded code to drive the servo sweep and read distance data, streaming real-time angle/distance readings to a processing interface.

ESP32 PWM Motor Control with Blynk IoT — *Embedded Systems / Motor Control* | ESP32, Blynk IoT, PWM/LEDC peripheral, Arduino IDE

- Built a Wi-Fi-controlled motor speed system using ESP32 PWM channels, exposing live speed control through the Blynk dashboard.
- Debugged and migrated PWM control code across ESP32 Arduino core API versions, resolving hardware-timer configuration issues.

Drone Motor Driver Research & Selection — *Robotics / Hardware Prototyping* | TB6612FNG, DRV8833, ESCs, brushed & brushless motors

- Researched and compared brushed vs. brushless motor driver options for a DIY drone build, evaluating TB6612FNG, DRV8833, and ESC-based approaches on a constrained budget.
- Explored software-based stabilization as a low-cost alternative to hardware gimbal stabilization, documenting trade-offs for future prototyping.

STM32 BlackPill Peripheral Bring-Up — *Embedded Systems / Microcontrollers* | STM32F411CEU6 (BlackPill), STM32CubeIDE, STM32CubeMX

- Set up the STM32CubeIDE toolchain and CubeMX peripheral configuration for a BlackPill board as a foundation for GPIO and timer-based experiments.

ESP32 Home Automation System — *IoT / Embedded Systems* | ESP32, Blynk, Relay module, Arduino IDE, Google Home, Alexa

- Built a Wi-Fi-controlled home automation system to safely toggle AC appliances remotely via relay modules and the Blynk app.
- Implemented multi-device control with real-time state feedback across a multi-device dashboard.

VLSI Design — Verilog RTL Modules — *Digital Design* | Xilinx Vivado, Verilog HDL

- Designed and simulated a 4-bit ALU, synchronous counter, and FSM-based traffic light controller.

EDUCATION

B.Tech — Electronics & Communication Engineering 2023 – 2027
Keshav Memorial College of Engineering, Hyderabad | CGPA: 6.05

Intermediate — MPC 2021 – 2023
Narayana Junior College | 85%

School — ICSE 2013 – 2021
Sri Sai Public School | 80%

CERTIFICATION

JIJNASA National Certificate — National Science Talent Competition

RELEVANT COURSEWORK

Embedded Systems • Microprocessors & Microcontrollers • Digital Electronics • VLSI Design • C Programming & Data Structures